

Let's Mine!

A guide to RapidMiner

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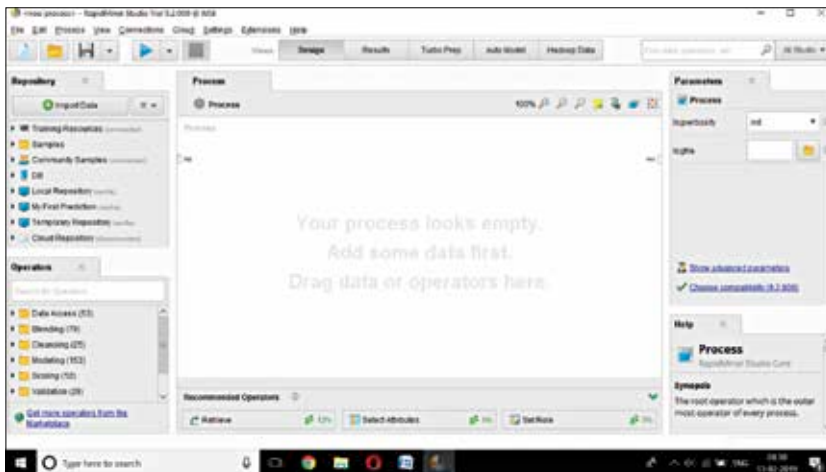
If you feel that data science or machine learning is your cup of tea, chances are you might have heard about RapidMiner. Quoted right from its official site, it is a “visual workflow designer that brings data and machine learning to the entire analytics team, from analysts to experts”

Marketplace. When it comes to databases, RapidMiner supports MySQL, PostgreSQL and Google BigQuery and the amazing part is that data can be queried and retrieved without you needing to write SQL (unless absolutely necessary)!

Let's now see what the RapidMiner studio has in store for us. A quick walkthrough the UI would help us.

THE WORKSPACE

The centre frame holds the 'Process' view where the actual work is being done like dragging in operators, connecting the ports, the results, etc.



This is what you're welcomed with when you open RapidMiner studio.

It boasts of over 1500 and more machine learning algorithms and functions that can be operated on data sets as per requirements. It supports drag and drop functionality like any other modern tool. It offers varieties of templates to suit use-cases which include customer defection, fraud detection, predictive analysis, so on and so forth. Databases, data-warehouses, cloud sources, social networks or simply data from local storage, RapidMiner can integrate with all of the above to retrieve data by means of extensions available at the RapidMiner

On the top left hand corner is the Repository view from which you get the data to work with. Rapid Miner provides us with a set of sample data. Database connections, data from cloud and even from local storage can be imported through this view. The bottom left hand corner has the operators view which has myriad operators to support operations we might want on our data. We have operations like retrieving, storing, performance classification, clustering, normalizing, detecting outliers, removing duplicate, predictive analysis, generating macros,

looping, executing SQL and R, creating directories, deleting files and what not. In essence, whatever modification you are doing on your data set, manually using an operator or via TurboPrep, in the back-end, it's always a process executing with operators chalked out in the exact sequence of your work.

The top right hand corner features the parameters view which comes in handy when you have dragged any operator from the operator view into the process view. For instance, if we drag a retrieve operator, which essentially retrieves data from the local storage, we need to mention the path from where it does. The path here is a parameter named as 'repository entry'. An interesting thing to note is that if you do know the name of the operator, you can just look it up using the search instead of travelling through the file structure to find it. In this case, we looked up retrieve.

The right bottom pane holds the 'Help' view where contextual information is provided regarding the operator which you have selected. It is a quick help and definitely not very descriptive. There are links to tutorials present in this view which can be clicked upon to get further elaboration.

Let's understand a bit on how rapid miner operates loading and saving of files. Let us first create a repository called 'My First Activity' and in it, create two folders called 'Data' and 'Processes', where the Data folder shall hold our datasets and the Processes folder shall hold our processes. RapidMiner processes have an extension of .rmp. Data in Rapid Miner are mostly Excel spreadsheets. Every row is called an example and every column, an attribute.

LOADING AND SAVING DATA

To keep things simpler, we are going to load the data from our local storage.

- **Without using a process** – This is simpler and can be thought of as a manual activity as opposed to using a process for the same. We click on import data in the repository view and load the file. In this case I have used